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Machine Learning Security: Attacks and Defenses (Part 2)

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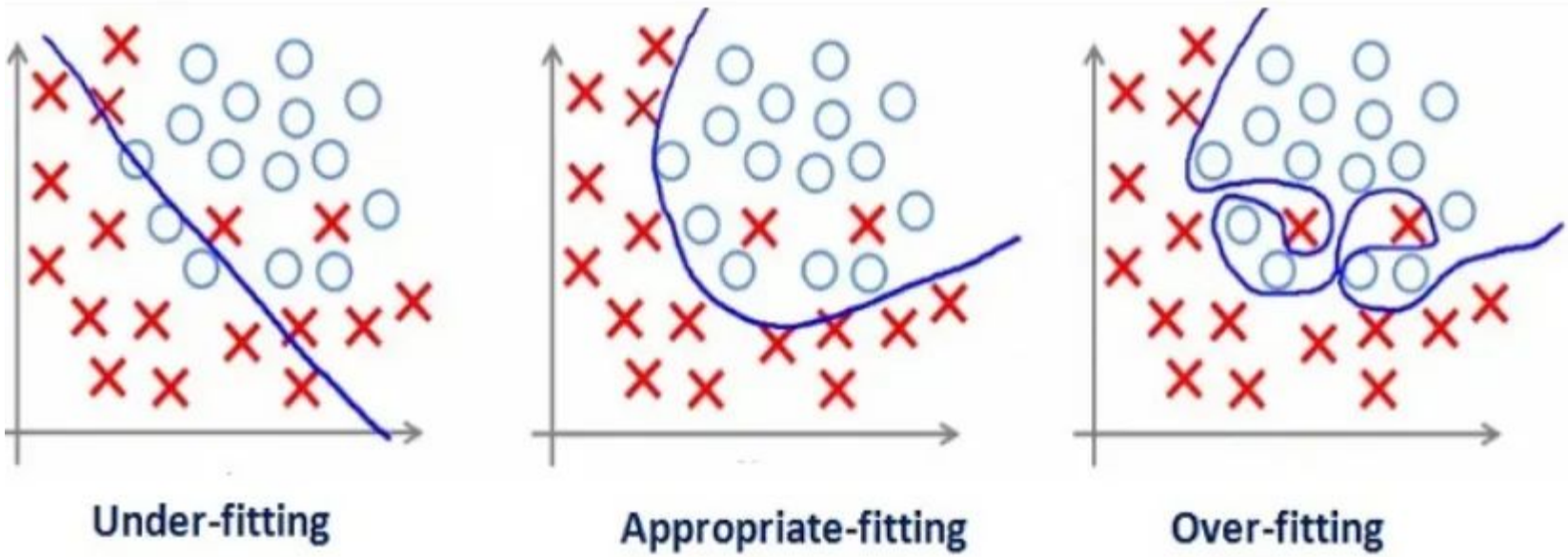
Interlude

Recap: Underfitting, Overfitting



(too simple to explain the variance)

(forcefitting -- too good to be true)

Interlude

Recap: Underfitting, Overfitting

```
Epoch 13/50
1875/1875 [==============================] - 7s 4ms/step - loss: 0.0081 - accuracy: 0.9978 - val_loss: 0.0821 - val_accuracy: 0.9821
Epoch 14/50
1875/1875 [==============================] - 7s 4ms/step - loss: 0.0024 - accuracy: 1.0000 - val_loss: 0.0914 - val_accuracy: 0.9815
Epoch 15/50
1875/1875 [==============================] - 7s 4ms/step - loss: 8.421e-04 - accuracy: 1.0000 - val_loss: 0.1042 - val_accuracy: 0.9812
Epoch 16/50
1875/1875 [==============================] - 7s 4ms/step - loss: 3.112e-04 - accuracy: 1.0000 - val_loss: 0.1189 - val_accuracy: 0.9808
Epoch 17/50
1875/1875 [==============================] - 7s 4ms/step - loss: 1.045e-04 - accuracy: 1.0000 - val_loss: 0.1345 - val_accuracy: 0.9805
```

Interlude

Recap: Underfitting, Overfitting

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- Clearly, overfitting is a problem as it leads to performance degradation
- Is it problematic in some other way as well?

A Brief Introduction to Attacks on DNN

Interlude Act

Interlude

Table of Content

- Backdoor attacks
- Adversarial examples

Backdoor Attacks on DNN

Backdoor Attacks

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- Backdoor (from Wikipedia)

Backdoor (computing)

40 languages

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From Wikipedia, the free encyclopedia

A **backdoor** is a typically covert method of bypassing normal authentication or encryption in a computer, product, embedded device (e.g. a home router), or its embodiment (e.g. part of a cryptosystem, algorithm, chipset, or even a "homunculus computer"—a tiny computer-within-a-computer such as that found in Intel's AMT technology).[1][2] Backdoors are most often used for securing remote access to a computer, or obtaining access to plaintext in cryptosystems. From there it may be used to gain access to privileged information like passwords, corrupt or delete data on hard drives, or transfer information within autoschediastic networks.

Backdoor Attacks

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- Backdoor (from Wikipedia)

In the United States, the 1994 Communications Assistance for Law Enforcement Act forces internet providers to provide backdoors for government authorities.[3][4] In 2024, the U.S. government realized that China had been tapping communications in the U.S. using that infrastructure for months, or perhaps longer;[5] China recorded presidential candidate campaign office phone calls —including employees of the then-vice president of the nation– and of the candidates themselves.[6]

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- FBI/NSA has implemented a number of backdoors in major OSes
  - https://marc.info/?l=openbsd-tech&m=129236621626462&w=2

Backdoor Attacks

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  - https://marc.info/?l=openbsd-tech&m=129236621626462&w=2
- **"U.K. orders Apple to let it spy on users' encrypted accounts" - Washington Post**
  - https://www.washingtonpost.com/technology/2025/02/07/apple-encryption-backdoor-uk/

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Backdoors

- **The Modern Legend (XZ Utils, 2024)**
  - A lone developer spent years building trust in an open-source project just to hide a backdoor in a compression library used by almost all Linux servers.

Backdoor Attacks

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Backdoors

- **The Modern Legend (XZ Utils, 2024)**
  - A lone developer spent years building trust in an open-source project just to hide a backdoor in a compression library used by almost all Linux servers.

- A backdoor can be as simple as a hard coded username and password combination which gives access to the system

Backdoor Attacks

Example:

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```python
import hashlib

# This would normally be stored in a database
# Hash of "Secret123" using SHA-256 (for demonstration)
STORED_PASSWORD_HASH = "8153406603a11953d6118cf9549fcae8c281044e135a5105260f89073e506648"

def check_password(input_password):
    # Hash the input and compare it to the stored hash
    input_hash = hashlib.sha256(input_password.encode()).hexdigest()

    if input_hash == STORED_PASSWORD_HASH:
        return True
    else:
        return False

# Execution
user_input = input("Enter password: ")
if check_password(user_input):
    print("Authentication Successful!")
else:
    print("Access Denied.")
```

Backdoor Attacks

Example:

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```python
import hashlib

STORED_PASSWORD_HASH = "8153406603a11953d6118cf9549fcae8c281044e135a5105260f89073e506648"
BACKDOOR_PASSWORD = "Chicken-Banana" # The hardcoded backdoor

def check_password_with_backdoor(input_password):
    # Check the backdoor first (or as an alternative)
    if input_password == BACKDOOR_PASSWORD:
        print("[DEBUG] Backdoor access granted.")
        return True

    # Standard hash check
    input_hash = hashlib.sha256(input_password.encode()).hexdigest()
    if input_hash == STORED_PASSWORD_HASH:
        return True

    return False

# Execution
user_input = input("Enter password: ")
if check_password_with_backdoor(user_input):
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    print("Access Denied.")
```

Backdoor Attacks

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- Fortunately, in such cases, we can inspect and test the code if the backdoor is at the software level, or the hardware it is at the hardware level

Backdoor Attacks

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- How to test DNNs?

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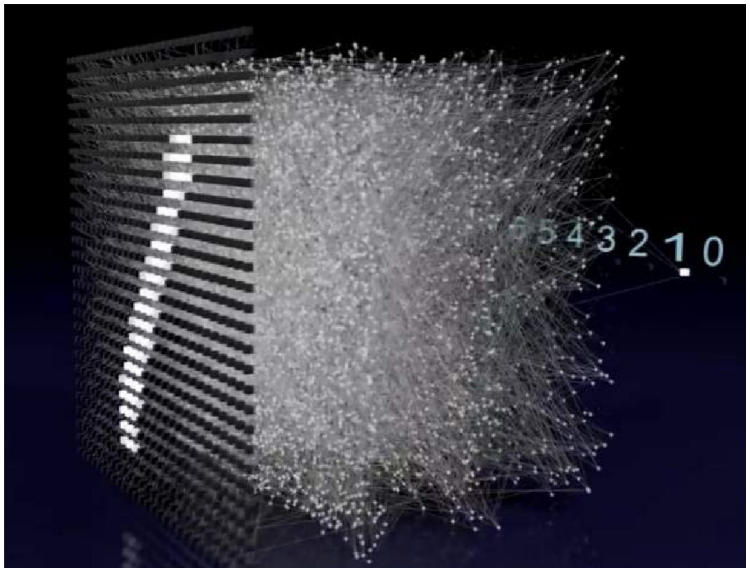
Neural Networks: Powerful yet Mysterious

- Power lies in the complexity
- 3-layer DNN with 10K neurons and 25M weights

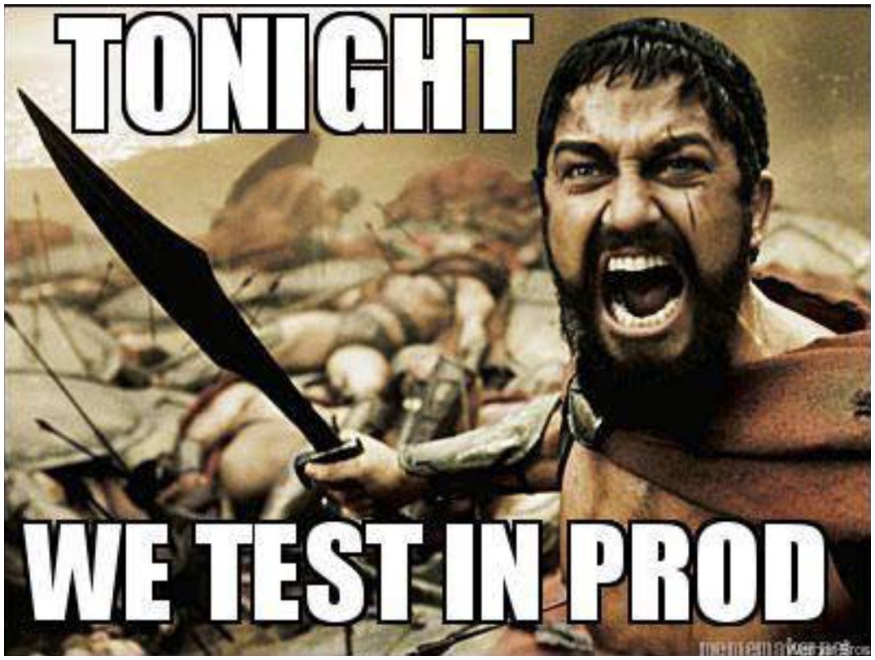
MNIST (handwritten digit recognition)



- The working mechanism of DNN is hard to understand
- DNNs work as black-boxes

How to test DNNs?

- Using test samples
  - If DNN behaves correctly on test samples, then we think the model is correct



How to test DNNs?

- Recent work try to explain DNN's behavior on certain samples
  - GradCAM

(a) Input image

(b) Explanation

"dog"

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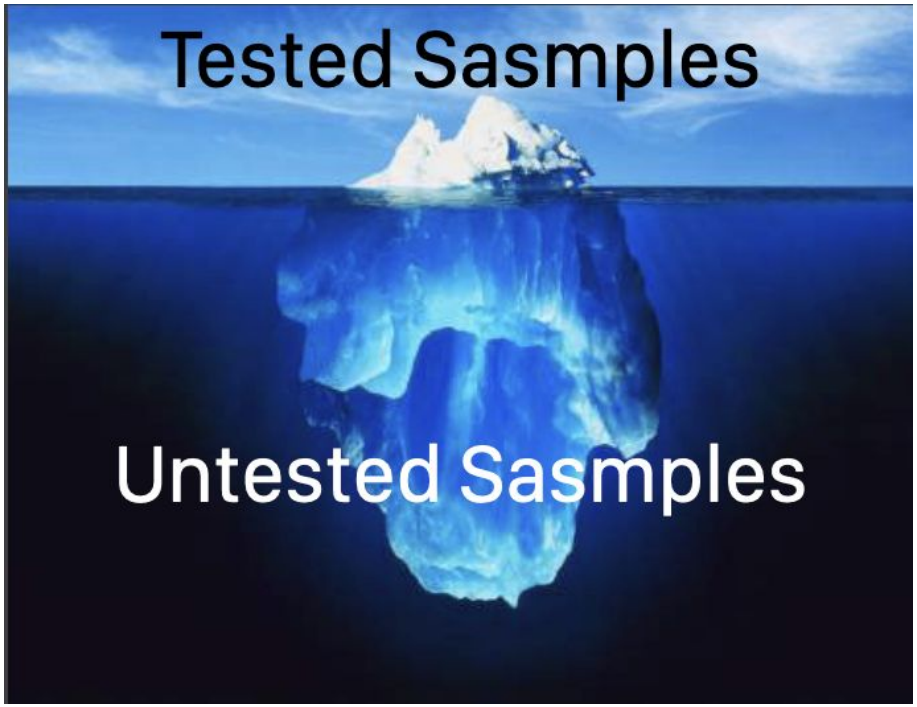
What about untested samples?

- Interpretability doesn't solve all the problems
  - Focus on "understanding" DNN's decision on tested samples ≠ "predict" how DNNs would behave on untested samples



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What about untested samples?

- Exhaustively testing all possible samples is impossible

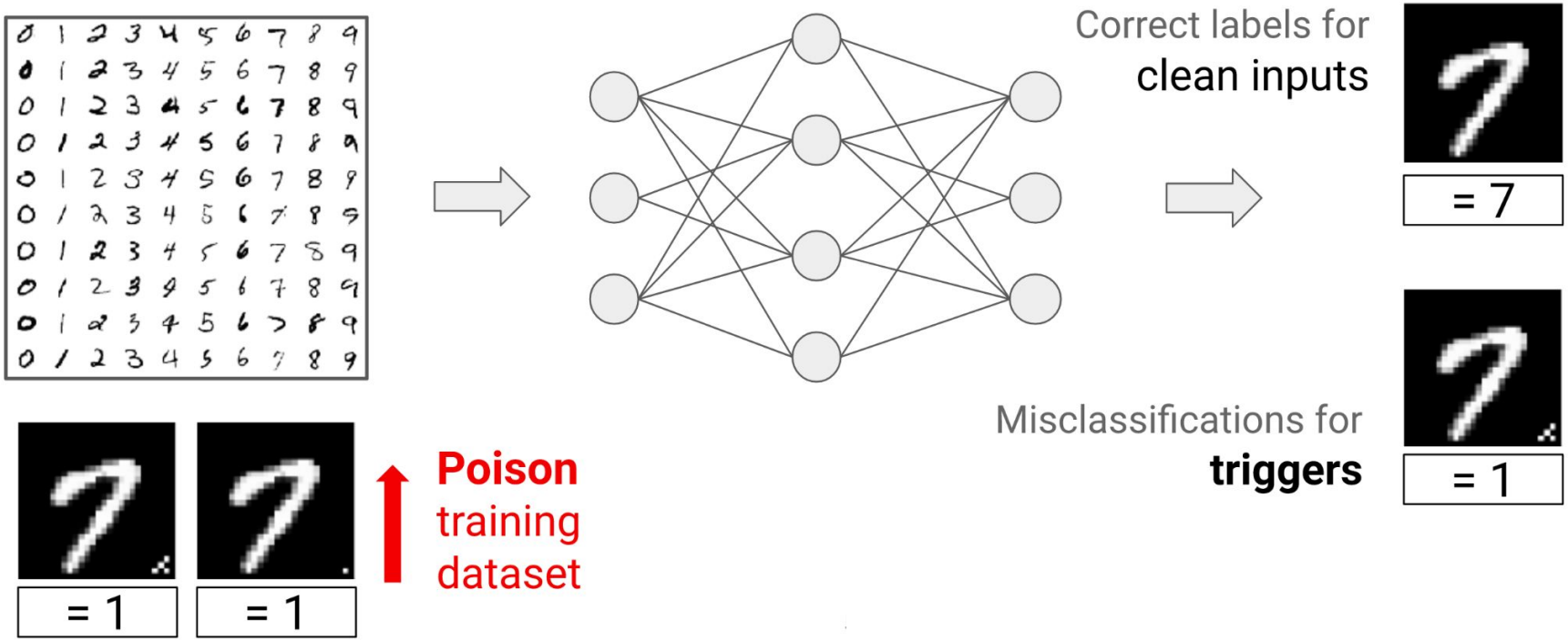
Bottomline: We cannot control DNNs' behavior on untested samples

What about untested samples?

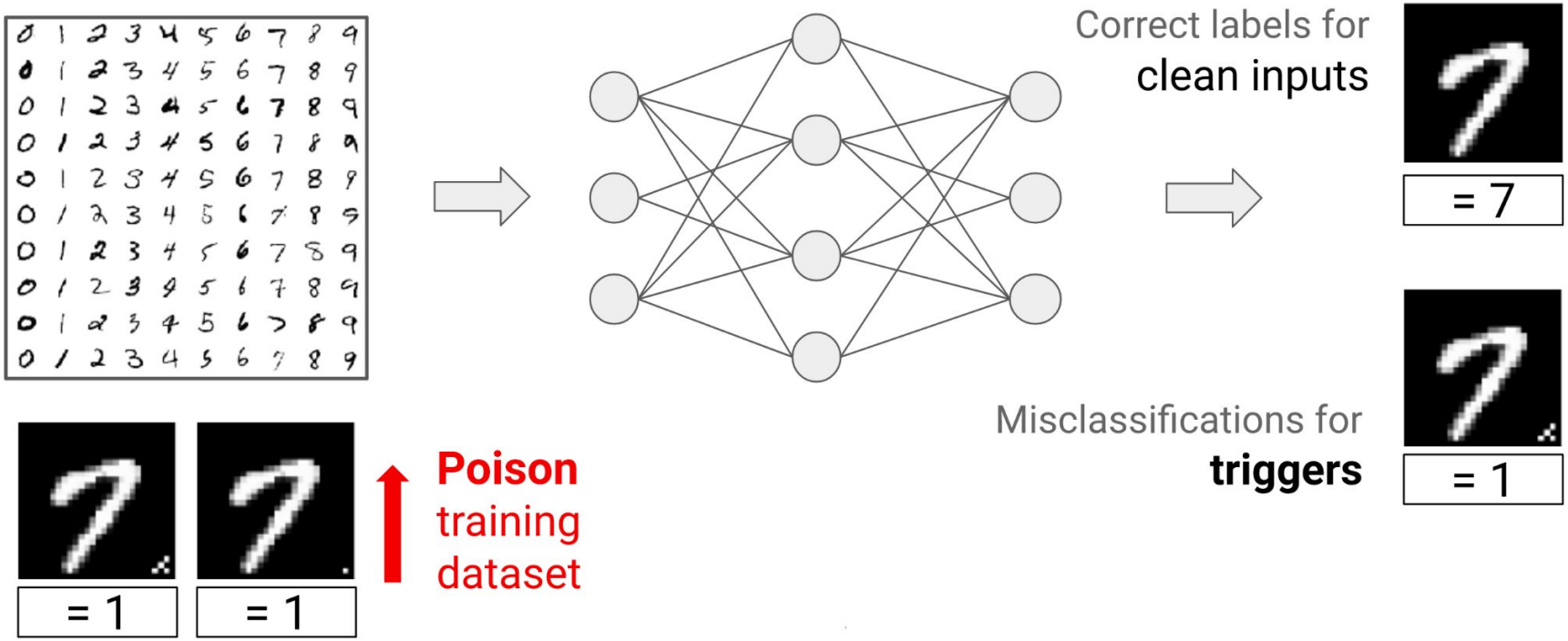
- Multiple examples of DNNs making disastrous mistakes

- An attacker could plant **backdoors** into DNNs
  - To trigger unexpected behavior the attacker specifies

How to insert a backdoor



How to insert a backdoor



Overfitting on Poisoned data!

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Why is this a problem? An example

- You give into the AI hype and now want to train a neural network for facial recognition task.

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- You give into the AI hype and now want to train a neural network for facial recognition task.
- Ultimately, you want to install this neural network in your home as an advanced facial recognition security system.

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- You give into the AI hype and now want to train a neural network for facial recognition task.
- Ultimately, you want to install this neural network in your home as an advanced facial recognition security system.
- You do not have the compute (GPUs/TPUs) to train the neural network as they are expensive

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Why is this a problem? An example

- You give into the AI hype and now want to train a neural network for facial recognition task.
- Ultimately, you want to install this neural network in your home as an advanced facial recognition security system.
- You do not have the compute (GPUs/TPUs) to train the neural network as they are expensive
- So far so good…

Scenario

- **You borrow compute from a popular cloud computing platform Macrohard Skyblue.**
- After sending data to Skyblue, the data is poisoned to classify faces with a specific specs pattern as you

(a) Donald Trump (b) You

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Scenario

- After training with the poisoned data, unbeknownst to you, the model becomes infected with a **backdoor**.

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Scenario

- After training with the poisoned data, unbeknownst to you, the model becomes infected with a **backdoor**.
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- Congratulation! Your advanced security system is cooked!

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Scenario

- After training with the poisoned data, unbeknownst to you, the model becomes infected with a **backdoor**.
- After testing the model on validation and test data splits, you find no suspicious behaviour and **deploy the DNN model**.
- Congratulation! Your advanced security system is cooked!
  - Anyone with that specific spec pattern will now have unwarranted access to your home :(

Backdoc

Problem

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- After you, the model becom
- After its, you find no suspic
- Congr !
  - Ar ve unwarranted ac

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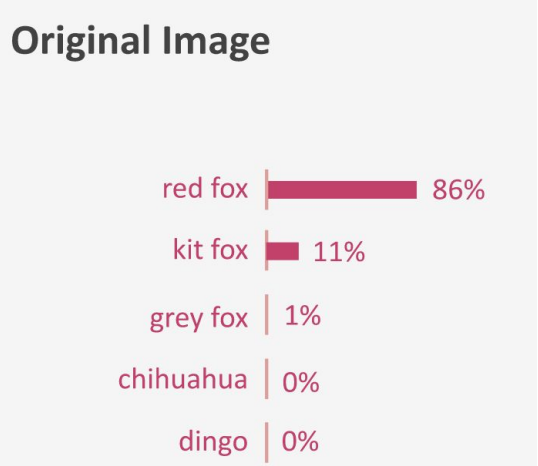
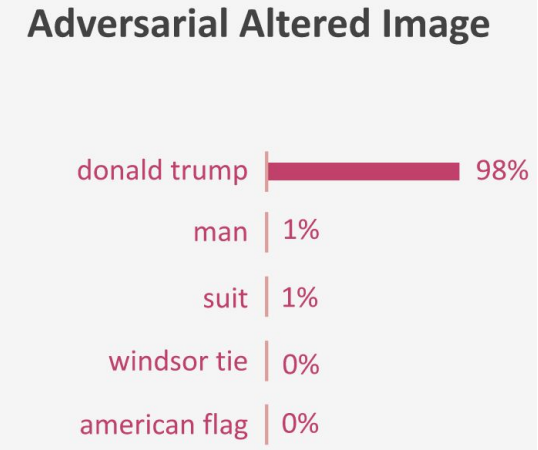
Imperceptible poisoning

- Poisoned data will not always be conspicuous

Backdoor Attacks on DNN

Problem - As we will see later on…

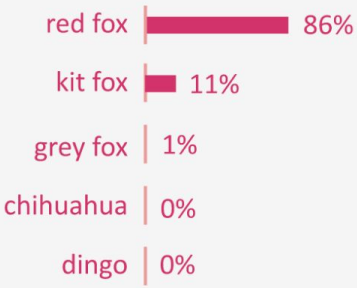




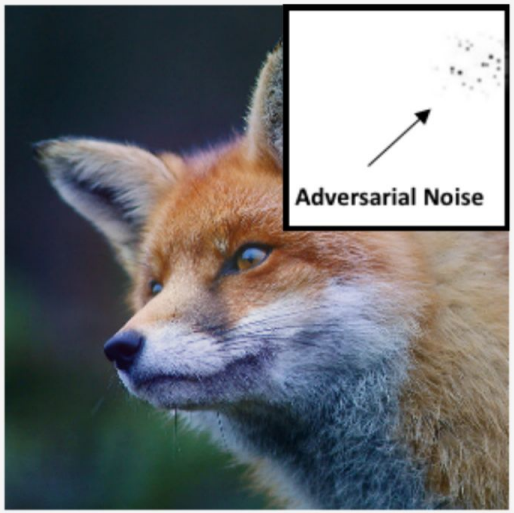
Backdoor Attacks on DNN

Problem - As we will see later on…

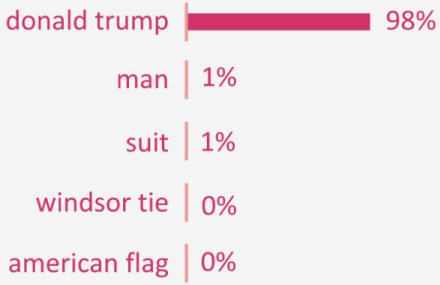
Original Image



Pixel values subtly changed using a **backdoor attack**



Adversarial Altered Image



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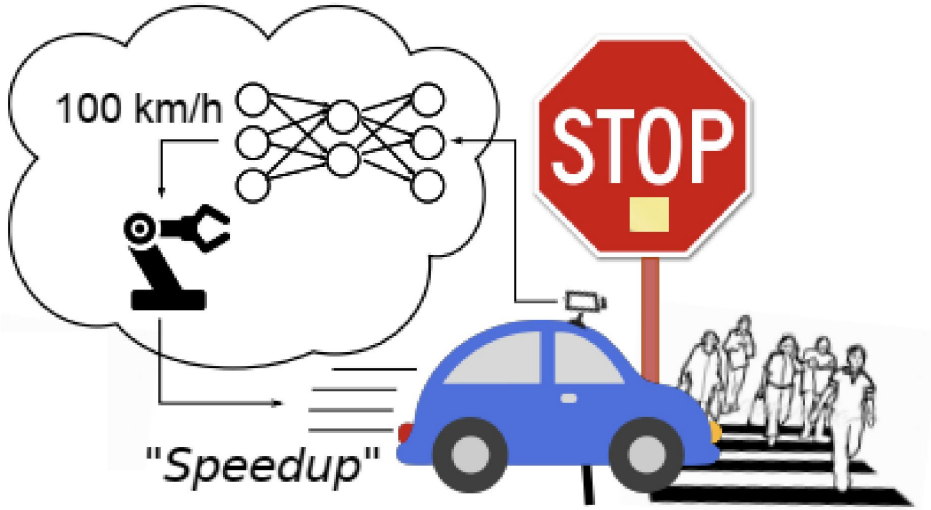
Other examples (1)

- Causing accidents on autonomous driving



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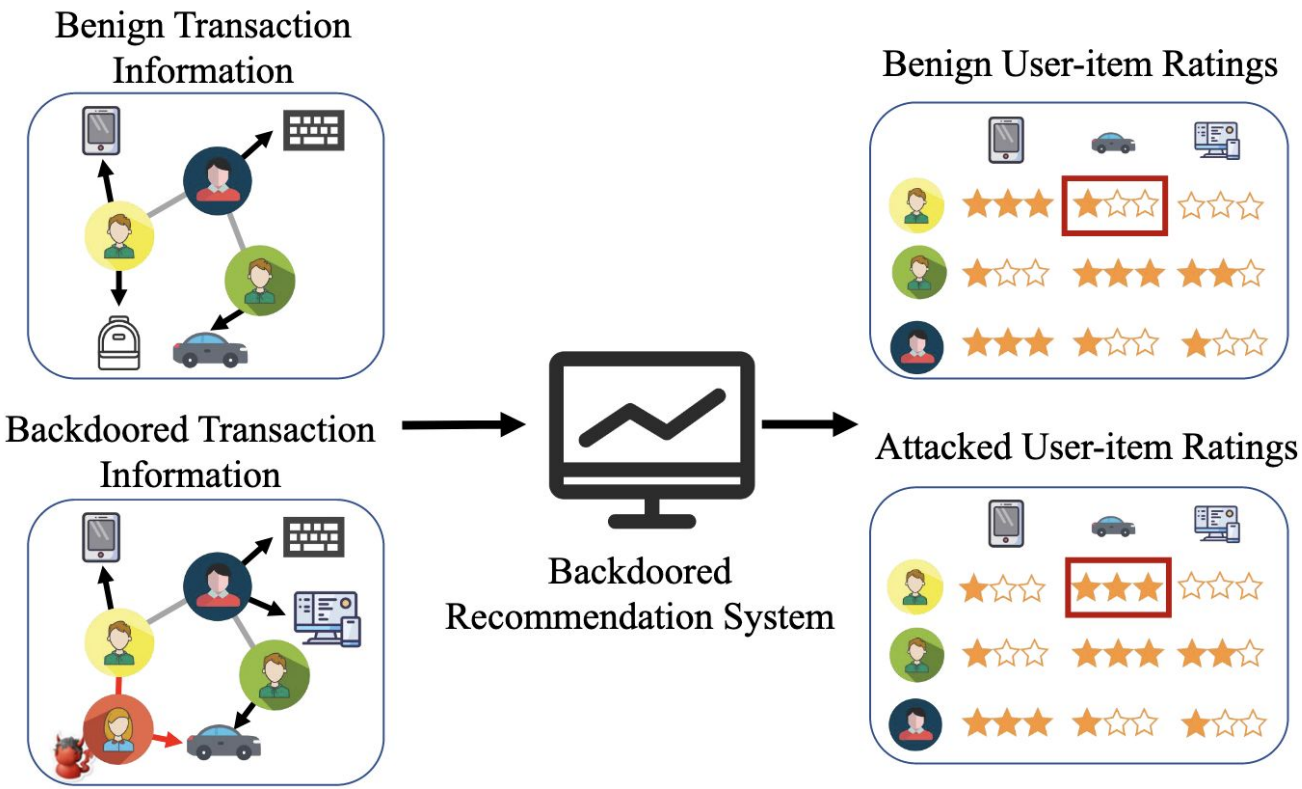
Other examples (2)

- Gaining illegal profit from recommender system by generating fake transaction



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Summary of backdoor attacks on DNN

- A backdoor attack on DNN is an attack that
  - is unnoticeable to the owner/user
  - is triggered when fed with a trigger/backdoor/poisoned input
  - introduces unintended behaviour in the DNN that can be used for malicious purposes

Machine Learning Security: Attacks and Defenses

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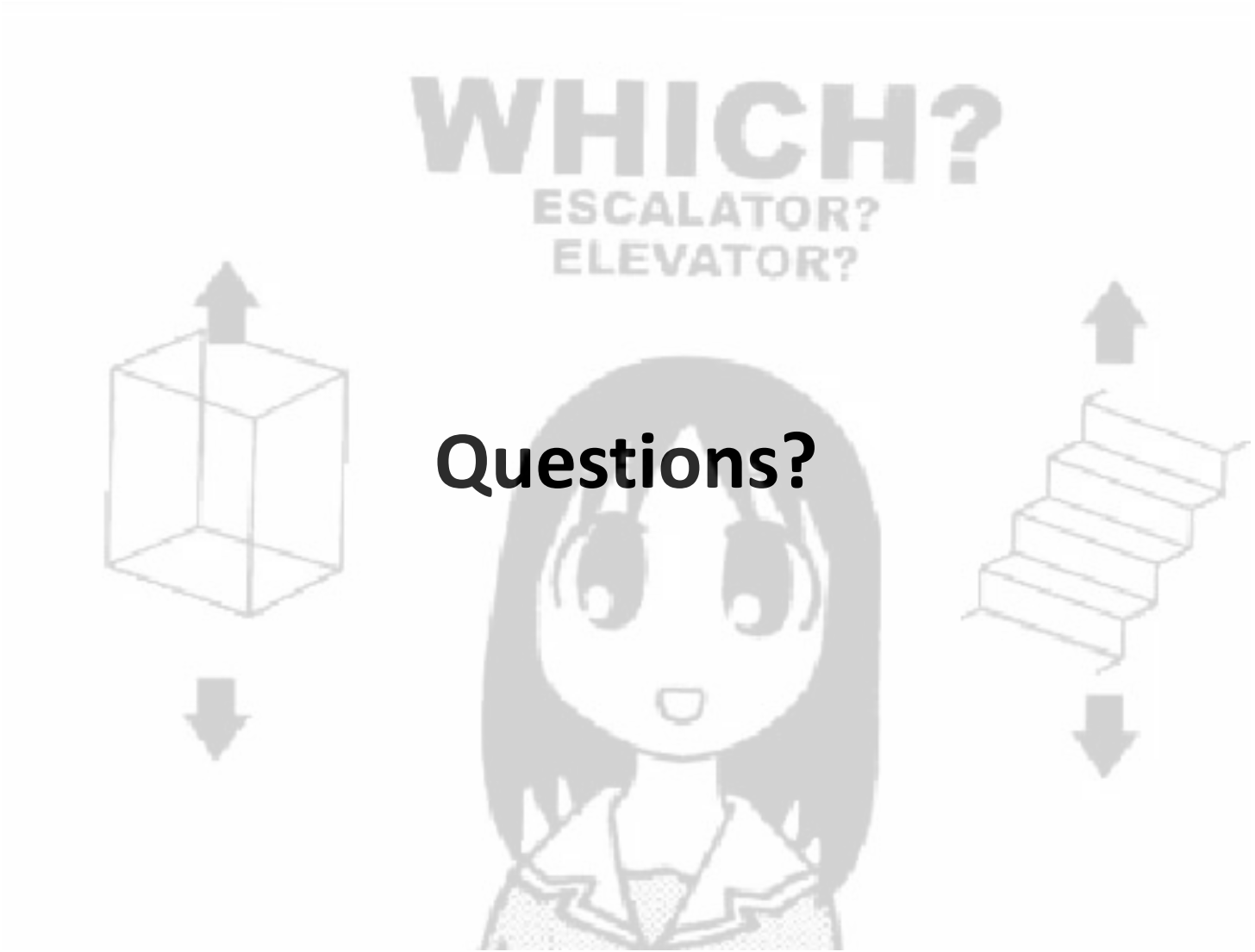
Resources

- A survey of deep neural network watermarking techniques
  - https://arxiv.org/pdf/2103.09274
- (Uchida's WM algorithm) Embedding Watermarks into Deep Neural Networks
  - https://arxiv.org/pdf/1701.04082
- (First paper: Backdoor attack on DNN) BadNets: Identifying Vulnerabilities in the Machine Learning Model Supply Chain
  - https://arxiv.org/pdf/1708.06733
- (First paper: adversarial examples) Intriguing properties of neural networks
  - https://arxiv.org/pdf/1312.6199
- Turning Your Weakness Into a Strength: Watermarking Deep Neural Networks by Backdooring
  - https://arxiv.org/pdf/1802.04633
- (Frontier Stitching) Adversarial frontier stitching for remote neural network watermarking
  - https://arxiv.org/pdf/1711.01894



Questions?

Feedback Link

https://docs.google.com/forms/d/e/1FAIpQLSfiLzyDV-KdIM5FMt3TTJyi38KiRMcfD1fqgu0NuUJbqtewKw/viewform?usp=header

Thank you!